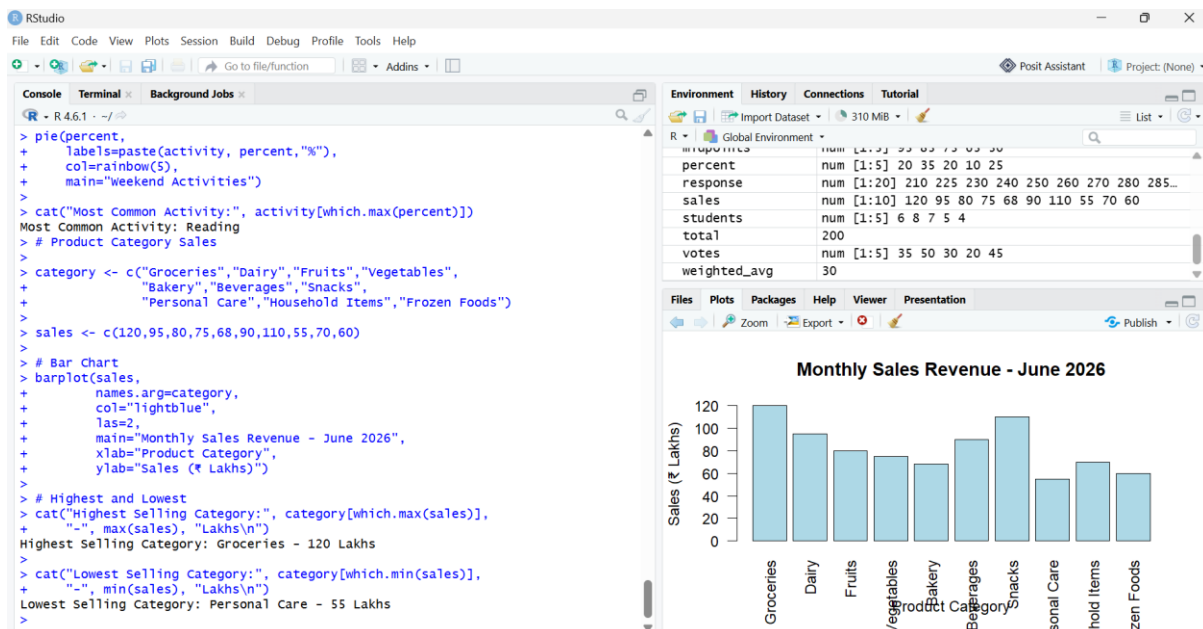


1.Bar Chart (Product Category Sales)

Code:

```
category <- c("Groceries","Dairy","Fruits","Vegetables",  
             "Bakery","Beverages","Snacks",  
             "Personal Care","Household Items","Frozen Foods")  
  
sales <- c(120,95,80,75,68,90,110,55,70,60)  
  
barplot(sales,  
        names.arg=category,  
        col="skyblue",  
        main="Monthly Sales Revenue - June 2026",  
        xlab="Product Categories",  
        ylab="Sales (₹ Lakhs)",  
        las=2)  
  
highest <- category[which.max(sales)]  
lowest <- category[which.min(sales)]  
  
cat("Highest Selling Category:", highest, "\n")  
cat("Lowest Selling Category:", lowest, "\n")
```



- barplot() creates the bar chart.
- which.max() identifies the highest sales category.
- which.min() identifies the lowest sales category.

- las=2 rotates category labels vertically.

Output

- Highest Selling Category = **Groceries (₹120 Lakhs)**
- Lowest Selling Category = **Personal Care (₹55 Lakhs)**

Observations

1. Groceries generated the highest sales revenue among all categories.
2. Personal Care recorded the lowest sales, indicating lower customer demand.

2.Histogram (Salary Packages)

Code:

```
salary <- c(3.2,4.5,5.0,6.8,7.2,8.5,
```

```
4.8,5.5,6.0,7.8,9.2,10.5)
```

```
hist(salary,
```

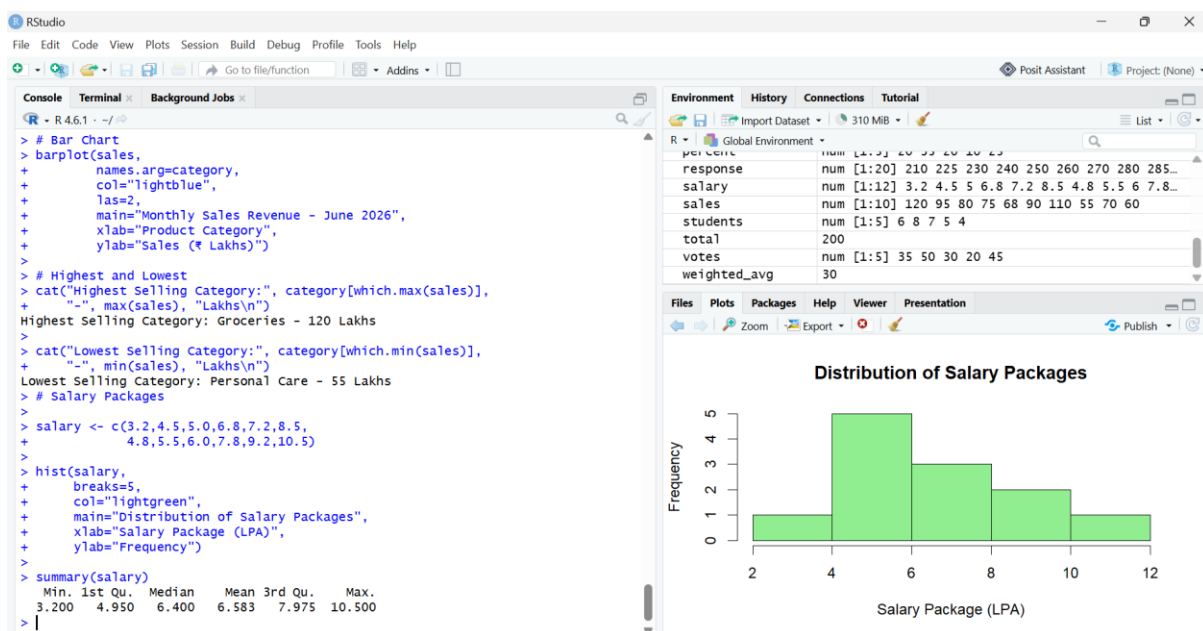
```
breaks=5,
```

```
col="lightgreen",
```

```
main="Distribution of Salary Packages",
```

```
xlab="Salary Package (LPA)",
```

```
ylab="Frequency")
```



- hist() creates the histogram.
- breaks=5 divides the salary range into five intervals.

- X-axis represents salary packages.
- Y-axis represents the number of students.

Analysis

- Salary packages range from **3.2 LPA to 10.5 LPA**.
- Most salaries lie between **4 and 8 LPA**.
- Higher salary packages (9.2 and 10.5 LPA) occur less frequently.
- No major outliers are observed, though **10.5 LPA** is the highest value.

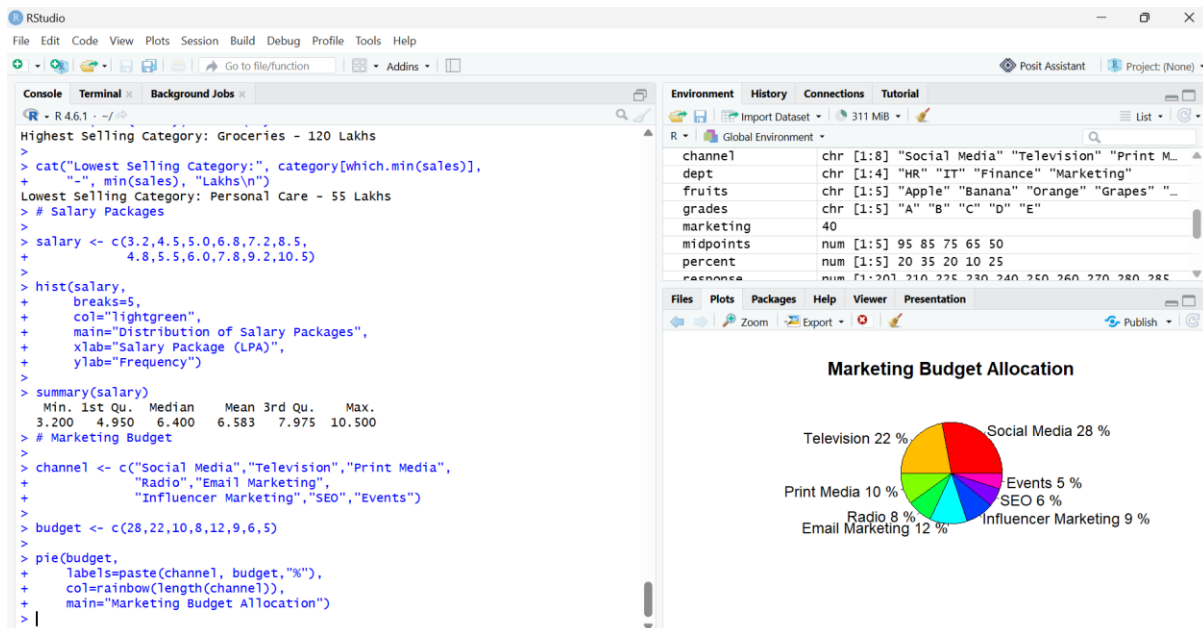
3.Pie Chart (Marketing Budget)

Code:

```
budget <- c(28,22,10,8,12,9,6,5)
channels <- c("Social Media",
              "Television",
              "Print Media",
              "Radio",
              "Email Marketing",
              "Influencer Marketing",
              "SEO",
              "Events")

pie(budget,
    labels=paste(channels,budget,"%"),
    col=rainbow(8),
    main="Marketing Budget Allocation")

highest <- channels[which.max(budget)]
cat("Highest Investment:", highest)
```



- pie() creates the pie chart.
- Each slice represents the percentage allocated to a marketing channel.
- which.max() finds the largest allocation.

Output

Highest Investment = **Social Media (28%)**

Distribution Analysis

- Social Media receives the highest investment.
- Television is the second-largest allocation (22%).
- Events receive the smallest allocation (5%).
- Most of the budget is concentrated in digital marketing channels.

4.Scatter Plot (Advertising vs Sales)

Code:

```
advertising <- c(5,8,10,12,15,18,20,22,25,28)
```

```
sales <- c(42,55,63,70,82,91,98,110,120,132)
```

```
plot(advertising,
```

```
    sales,
```

```
    pch=19,
```

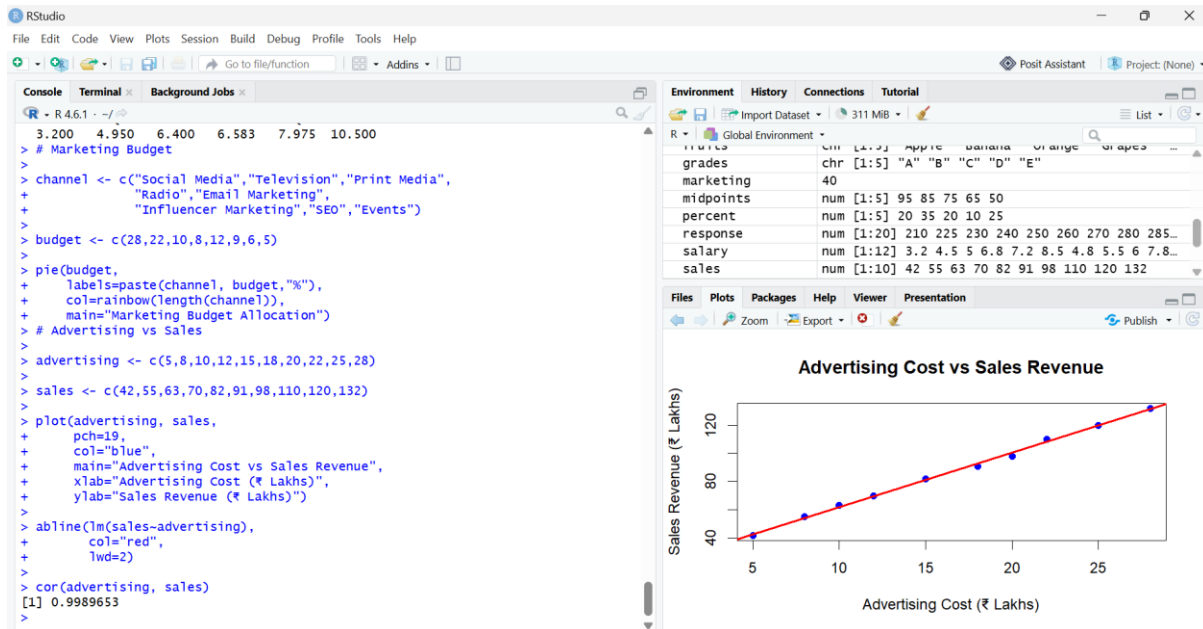
```
    col="blue",
```

```
    main="Advertising Cost vs Sales Revenue",
```

```

xlab="Advertising Cost (₹ Lakhs)",
ylab="Sales Revenue (₹ Lakhs)")
abline(lm(sales~advertising), col="red", lwd=2)
cor(advertising,sales)

```



- `plot()` creates the scatter plot.
- `abline()` adds the regression line.
- `cor()` calculates the correlation coefficient.

Analysis

- Sales increase as advertising cost increases.
- The points lie close to the regression line.
- Correlation is close to +1, indicating a **strong positive relationship**.

5. Line Chart (Electricity Generation)

Code:

```

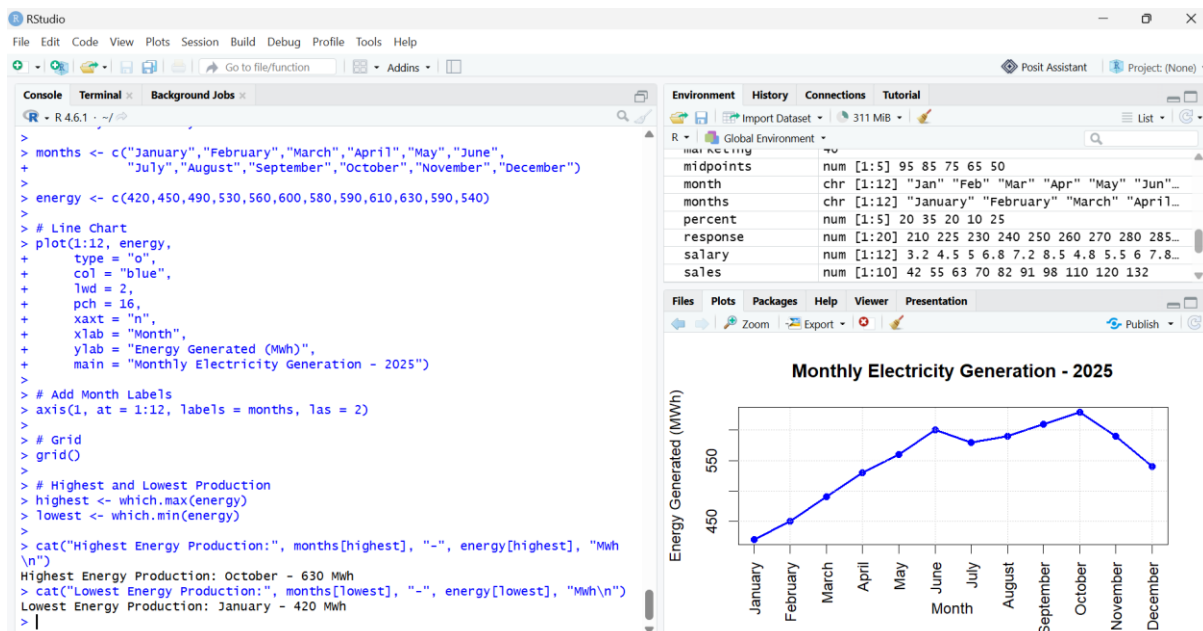
months <- c("Jan","Feb","Mar","Apr","May","Jun",
            "Jul","Aug","Sep","Oct","Nov","Dec")
energy <- c(420,450,490,530,560,600,
            580,590,610,630,590,540)
plot(months,
     energy,

```

```

type="o",
col="blue",
pch=16,
lwd=2,
main="Monthly Solar Energy Generation - 2025",
xlab="Month",
ylab="Energy Generated (MWh)")
highest <- months[which.max(energy)]
lowest <- months[which.min(energy)]
cat("Highest Production:", highest, "\n")
cat("Lowest Production:", lowest)

```



- `type="o"` displays both lines and points.
- `pch=16` creates filled circle markers.
- `which.max()` identifies the highest production month.
- `which.min()` identifies the lowest production month.

Output

Highest Production = **October (630 MWh)**

Lowest Production = **January (420 MWh)**